

AYE
TECH HUB

• HVAC ENGINEERING • LESSON 05

Heating Systems

Boilers, Heat Pumps, UFH & Radiators

Lesson 05 of 30 | HVAC Engineering Program

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- Boiler types: gas, oil, biomass, electric and condensing
- How heat pumps achieve COP 3-5 for heating
- Underfloor heating: wet vs dry systems, pipe spacing, controls
- Radiator sizing and low-temperature heating circuit design
- Heat loss calculation and heating system selection guide

Heating Systems — Principles and Plant Types

A heating system transfers thermal energy from a heat source to the occupied space. Every system has three parts: **heat generation** (boiler, heat pump), **heat distribution** (pipes, pumps, controls), and **heat emission** (radiators, underfloor, fan coils). Selecting the right combination for the building type, climate, and energy source is the core task of heating design.

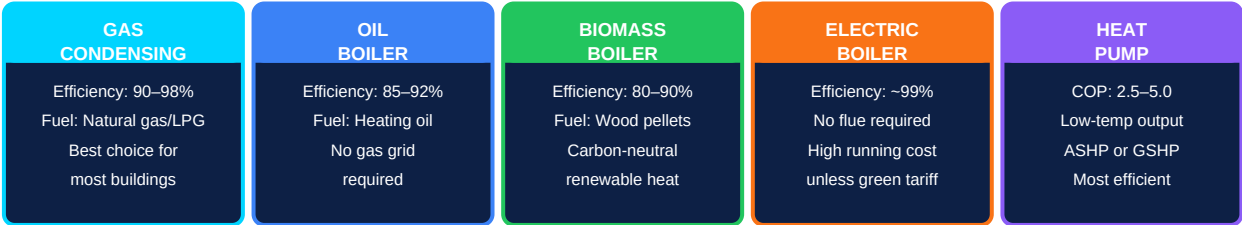


Figure 1 — Heating Plant Types and Key Characteristics

Boiler Type	Efficiency	Flow / Return Temp	Flue Required	Best Application
Condensing gas (combi)	90–98% (SEDBUK A)	70°C flow / 50°C return (or LT 55/45°C)	Yes — balanced flue	Domestic, small commercial; instant hot water
Condensing gas (system)	90–98%	70°C flow / 50°C return	Yes	Larger domestic; separate hot water cylinder
Oil condensing	85–93%	70°C flow / 50°C return	Yes — higher clearance	Rural buildings without gas grid connection
Biomass (wood pellet)	80–92%	70°C flow / 50°C return	Yes — large flue	Carbon-neutral heating; rural schools, community halls
Electric flow boiler	~100% (point of use)	70°C flow / 50°C return	No flue needed	Small flats, apartments; paired with heat pump tariff

Condensing boiler operation: A condensing boiler recovers latent heat from flue gases by cooling them below the dew point (~55°C). This condensation releases an extra 10–12% of energy that non-condensing boilers waste. To condense efficiently, return water temperature must be below 55°C — which is why low-temperature systems (underfloor heating) maximise efficiency.

Heat Pumps — The Most Efficient Heating Technology

A heat pump does not generate heat — it **moves heat** from a low-temperature source (outdoor air, ground, or water) to a higher-temperature heating circuit. Because it moves heat rather than creating it, a heat pump can deliver 3–5 units of heat energy for every 1 unit of electrical energy consumed. This is expressed as the **Coefficient of Performance (COP)**.

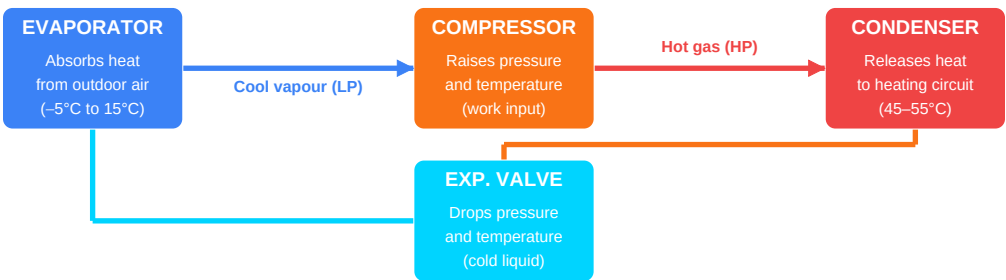


Figure 2 — Air-Source Heat Pump Cycle: $COP = \text{Heat Output} \div \text{Work Input}$

Heat Pump Type	Heat Source	Typical COP	Installation	Best For
Air Source HP (ASHP)	Outdoor ambient air (−15°C to 35°C)	2.5–4.5 (COP drops in very cold weather)	Outdoor unit + indoor hydronic system	Residential retrofit, moderate climates
Ground Source HP (GSHP)	Ground (boreholes or horizontal loops at 8–12°C constant)	3.5–5.5 (stable COP year-round)	Borehole drilling or land excavation	New build, large sites, cold climates
Water Source HP (WSHP)	River, lake, or aquifer (10–20°C)	4.0–6.0 (highest COP)	Water abstraction licence + heat exchanger	Buildings near water bodies, district energy
Exhaust Air HP	Warm extract air from ventilation (18–22°C)	3.0–4.0	Integrated with MVHR unit	High-insulation buildings with MVHR system

Parameter	ASHP Design Value	Why It Matters
Minimum outdoor design temp	Depends on climate: UK: −5°C, Scandinavia: −20°C	Heat pump must still meet building heat demand at coldest design day
Flow temperature	45–55°C (LT), up to 65°C (high-temp HP)	Lower flow temp → higher COP; LT emitters (UFH) essential for efficiency
Defrost cycle	Automatic at −5°C to +5°C outdoor temp	Evaporator frosts; defrost uses energy and interrupts heating output
Seasonal COP (SCOP)	EU A++ ASHP: SCOP ≥ 3.1 (Tj climate)	More realistic than laboratory COP; used for energy label ratings
Buffer vessel	50–200 L recommended for most ASHPs	Prevents short-cycling; stabilises flow temperature; protects compressor

ASHP vs gas boiler running cost: At UK electricity 28p/kWh and gas 7p/kWh, a COP 3.0 ASHP costs $28 \div 3 = 9.3\text{p/kWh}$ of heat. A 92% gas boiler costs $7 \div 0.92 = 7.6\text{p/kWh}$. ASHP becomes cheaper when COP exceeds electricity-to-gas price ratio (currently ~4.0 in UK). In countries with cheap electricity or expensive gas, ASHPs win easily.

Heat Emitters, Heat Loss and System Selection

The heat emitter is the final link between the heating system and the occupant. The choice of emitter determines the required flow temperature — which in turn determines the system efficiency. Low-temperature emitters (underfloor heating) are essential for maximising heat pump efficiency.

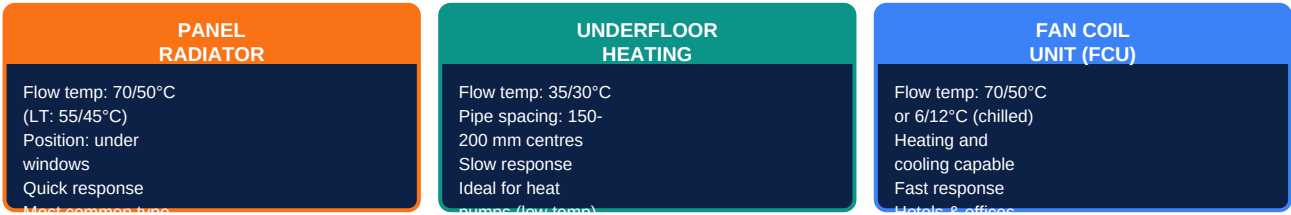


Figure 3 — Heating Emitter Types and Operating Temperatures

Emitter	Design Flow / Return	Response Time	Suitable with Heat Pump	Typical Use
Steel panel radiator (HT)	70°C / 50°C	Fast (15–30 min)	Limited (needs high-temp HP)	Standard domestic retrofit with gas boiler
Low-temp panel radiator (oversized)	55°C / 45°C	Moderate (20–40 min)	Yes (standard ASHP)	Heat pump retrofit where UFH not possible
Underfloor heating (wet)	35°C / 30°C	Slow (2–4 hrs)	Excellent (ideal pairing)	New build, extensions, ground-floor slabs
Underfloor heating (electric)	Direct electric (no circuit)	Moderate (30–60 min)	N/A (not hydronic)	Bathrooms, small areas — high running cost
Fan coil unit (4-pipe)	70/50°C (heat) 6/12°C (cool)	Fast (5–10 min)	Yes (with heat pump)	Hotels, offices, commercial — heating and cooling

Heat Loss Calculation and System Sizing

Step	Calculation	Typical Values
1. Design temperature difference	$\Delta T = T_{\text{indoor}} - T_{\text{outdoor_design}}$	UK: 21°C indoor – (–5°C) outdoor = 26 K Ethiopia highlands: 18°C – 5°C = 13 K
2. Fabric heat loss	$Q_{\text{fabric}} = U \times A \times \Delta T$ (W) Sum over all walls, roof, floor, glazing	Well-insulated wall: $U=0.25$ W/m²K Double glazing: $U=1.8$ W/m²K
3. Ventilation heat loss	$Q_{\text{vent}} = 0.33 \times N \times V \times \Delta T$ (W) N =air changes/hr, V =room volume	Typical domestic: $N=0.5$ ACH Q_{vent} usually 20–30% of total heat loss
4. Total design heat load	$Q_{\text{total}} = Q_{\text{fabric}} + Q_{\text{vent}}$ (W or kW)	Typical UK semi: 6–10 kW Well-insulated new build: 3–5 kW
5. Boiler / HP sizing	Select plant at 100–110% of design heat load (do not oversize — causes cycling)	Gas boiler: size to design load ASHP: size to 85% — backup immersion for peaks
6. Radiator / emitter sizing	Emitter output = Q_{room} / number of emitters At design delta T	Radiator output data in manufacturer schedules UFH: 100 W/m² typical output

What comes next — HVAC-006: Ventilation Principles — natural ventilation, mechanical extract, MVHR (mechanical ventilation with heat recovery), ventilation rates, CO₂ monitoring, and designing compliant ventilation for residential and commercial buildings.